Template Document

Academic Support Agent

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1. Question summary/restatement can go here.

   (a) Summary of the first part of the question can go here.

   Down here is then where you can place your solution. Math can be entered using the equation environment like this

   $$\vec{\mathbf{r}} = \vec{\mathbf{r}}_0 + \vec{\mathbf{v}}_0 t + \frac{1}{2}\vec{\mathbf{a}} t^2 \tag{1}$$

   If you then where working in say the x-direction and had some numbers

   $$\begin{aligned} x &= x_0 + v_{x0}t + \tfrac{1}{2}a_x t^2 \\ &= 1.2 \text{ m} + (4.0 \text{ m/s})(3.0 \text{ s}) + \tfrac{1}{2}(-1.0 \text{ m/s}^2)(3.0 \text{ s})^2 \\ &= \boxed{8.7 \text{ m}} \end{aligned} \tag{2}$$

   Here we used an array so that we could have multiple lines in a single equation environment. The `@{~=~}` sets things up so that when we type a & it will put in a space, then and equals sign and another space. The r means to take what is on the left of the & and align it to the right and the l then means to left align what follows the &.

   (b) When you get to the next part, just add a `\item` to get the appropriate label. Also, if you don't like all the equation numbers, you can use the following to have the equation with no number

   $$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}$$

   (c) For more details on putting math into LaTeX documents you can see this page on Overleaf.

2. We you get to the next problem, just make sure to end the enumerate for the parts of the previous problem and then add another item.

   (a) Use a nested enumerate environment to label the parts of the next problem.

   (b) For a quick and broad overview of how to create documents in LaTeX see this quick tutorial from Overleaf.